1.2.41

Signal Processing – Discrete

MN 2007

Question

The solution of

$$ye^x\,dx + (4y + e^x)\,dy = 0, \quad y(0) = -1 \tag{1}$$

is

a) $ye^x + 2y^2 - 1 = 0$

b) $e^x + y^2 - 2 = 0$

c) $ye^x - y^2 = 0$

d) $ye^x + y^2 - 1 = 0$

Solution

Rearrange by dividing throughout by dy:

$$ye^x \frac{dx}{dy} + 4y + e^x = 0 \tag{2}$$

$$ye^x \frac{dx}{dy} + e^x = -4y \tag{3}$$

Observe the product rule for differentiation with respect to y:

$$\frac{d}{dy}(ye^x) = e^x + ye^x \frac{dx}{dy} \tag{4}$$

The left-hand side is exactly $\frac{d}{dy}(ye^x)$, so:

$$\frac{d}{dy}(ye^x) = -4y \tag{5}$$

Solution

Integrating both sides with respect to y:

$$\int \frac{d}{dy}\left(ye^x\right)\, dy = \int -4y\, dy \tag{6}$$

$$ye^x = -2y^2 + C \tag{7}$$

Applying the initial condition $y(0) = -1$, i.e. $x = 0$ when $y = -1$:

$$(-1)\, e^0 = -2\,(-1)^2 + C \tag{8}$$

$$-1 = -2 + C \tag{9}$$

$$C = 1 \tag{10}$$

Solution

Substituting $C = 1$ into (7):

$$ye^x = -2y^2 + 1 \tag{11}$$

$$\boxed{ye^x + 2y^2 - 1 = 0} \tag{12}$$

Verification: At $y(0) = -1$,

$$(-1)\,e^0 + 2\,(-1)^2 - 1 = -1 + 2 - 1 = 0 \quad \checkmark \tag{13}$$

Answer

Option (a): $ye^x + 2y^2 - 1 = 0$

How the Numerical (odeint) Plot is Obtained

The implicit ODE $ye^x\,dx + (4y + e^x)\,dy = 0$ is rewritten as an explicit $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{-ye^x}{4y + e^x} \tag{14}$$

`scipy.integrate.odeint` solves this using the **LSODA** algorithm:

- Starts at the initial condition $y(0) = -1$
- At each step x_k, evaluates $\frac{dy}{dx}\big|_{x_k,\,y_k}$ from (14)
- Advances: $y_{k+1} \approx y_k + h \cdot \frac{dy}{dx}\big|_{x_k,\,y_k}$ using adaptive Runge-Kutta / Adams methods
- Automatically shrinks step size h where curve bends sharply

Note

Accuracy degrades when $4y + e^x \to 0$ (denominator $\to 0$, slope $\to \infty$).

How the Analytical Plot is Obtained

The implicit solution $ye^x + 2y^2 - 1 = 0$ is a quadratic in y:

$$2y^2 + e^x \cdot y - 1 = 0 \tag{15}$$

Applying the quadratic formula with $A = 2,\ B = e^x,\ C = -1$:

$$y = \frac{-e^x \pm \sqrt{e^{2x} + 8}}{4} \tag{16}$$

Select the $-$ branch to match $y(0) = -1$:

$$y\big|_{x=0} = \frac{-1 - \sqrt{1+8}}{4} = \frac{-1-3}{4} = -1 \quad \checkmark \tag{17}$$

Closed-form used for plotting

$$y(x) = \frac{-e^x - \sqrt{e^{2x} + 8}}{4}$$

This is exact for all x — no accumulation of numerical error.

Plot

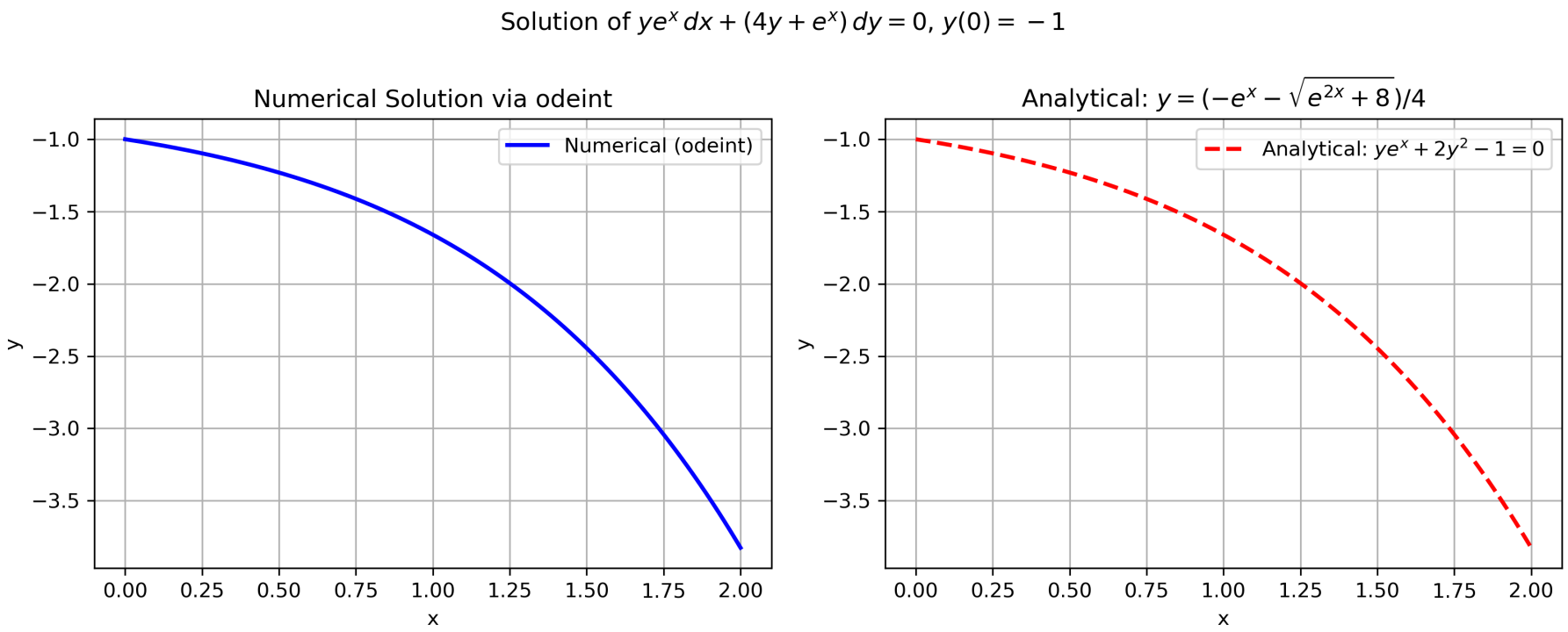

Figure: Left: Numerical via `odeint`. Right: Analytical closed-form.

Python Code: Setup and Numerical Solution

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint


def ode(y, x):
    ex = np.exp(x)
    denom = 4*y + ex
    if abs(denom) < 1e-10:
        return 0.0
    return -y * ex / denom

x = np.linspace(0, 2, 500)
sol = odeint(ode, [-1.0], x)
```

Python Code: Analytical Solution

```
def y_analytical(x_vals):
    ex = np.exp(x_vals)
    return (-ex - np.sqrt(ex**2 + 8)) / 4

y_anal = y_analytical(x)

print(f"Analytical y(0) = {y_analytical(np.array([0.0]))[0]:.4f}"
    )
```

Python Code: Separate Plots

```
fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

axes[0].plot(x, sol, 'b-', linewidth=2, label='Numerical (odeint)
    ')
axes[0].set_title('Numerical Solution via odeint')
axes[0].set_xlabel('x'); axes[0].set_ylabel('y')
axes[0].legend(); axes[0].grid(True)

axes[1].plot(x, y_anal, 'r--', linewidth=2,
            label=r'Analytical: $ye^x+2y^2-1=0$')
axes[1].set_title(r'Analytical: $y=(-e^x-\sqrt{e^{2x}+8})/4$')
axes[1].set_xlabel('x'); axes[1].set_ylabel('y')
axes[1].legend(); axes[1].grid(True)

plt.suptitle(r'Solution of $ye^x\,dx+(4y+e^x)\,dy=0$, $y(0)=-1$')
plt.tight_layout()
plt.savefig("image.png", dpi=300)
plt.show()
```